Final Year Project Summary

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Degree	Bachelor of Science in Artificial Intelligence
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Project Title	AI Virtual Makeup Try-On
Project Type	Group Final Year Project (3 members)

Project Overview

The project developed a real-time virtual makeup application that allows users to preview cosmetic products through a live camera feed. The system was designed as a practical digital alternative to physical makeup testing, with emphasis on convenience, hygiene, personalization, and real-time interaction. It supports virtual application of foundation, lipstick, eyeliner, eyeshadow, and eyebrow colour.

Methodology and Technical Implementation

The project followed an Agile development approach, with work divided into iterative stages covering facial detection, makeup overlay implementation, interface development, integration, testing, and refinement. The final desktop application was implemented mainly in Python. MediaPipe Face Mesh was used to detect and track 468 facial landmarks in real time, OpenCV handled image processing and makeup-overlay rendering, and PySide6 was used to build the graphical user interface. The system maps makeup effects to the correct facial regions and updates them dynamically as the user moves, smiles, blinks, or changes head position.

Testing and Evaluation

The application was evaluated through functional, usability, and performance testing on different hardware configurations. Testing focused on facial-landmark accuracy, overlay alignment, response time, rendering stability, product intensity controls, and multi-product use. Identified issues such as slight overlay misalignment, blinking-related flicker, and performance bottlenecks were addressed through landmark smoothing, improved coordinate mapping, blending adjustments, and rendering optimization.

Outcome

The final result was a functional desktop prototype capable of real-time facial tracking, multiple simultaneous makeup overlays, adjustable intensity, and responsive user interaction. The project demonstrated how computer vision, image processing, and AI-oriented techniques can be combined to solve a practical consumer problem and provided a strong basis for future extensions such as improved skin-tone analysis, recommendation systems, 3D face modelling, and mobile or web deployment.

Individual Contribution

Although the project was completed as a three-member group project, I carried out the majority of the core technical and project work. My main contributions included system architecture and methodology, Python implementation, MediaPipe-based facial landmark detection, OpenCV-based overlay rendering, PySide6 interface integration, landmark mapping and blending, testing and debugging, performance optimization, research and literature work, and preparation of the final project documentation and report.